



K-IDS: An Intrusion Detection System in Kubernetes Clusters Using eBPF and Machine Learning

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Introduction

- Kubernetes environments are dynamic and introduce complex security challenges, especially in containerized and multi-tenant systems **(Hussain et al., 2025)**.
- Machine Learning-based IDS can detect anomaly patterns and emerging threats, but relies heavily on high-quality data for accurate detection **(Putro et al., 2025)**.
- eBPF enables real-time, low-overhead, kernel-level monitoring of system and network behavior **(Hussain et al., 2025)**.
- An IDS for Kubernetes that **combines eBPF (observability) and Machine Learning (detection) to deliver real-time, fine-grained, and adaptive intrusion detection.**



eBPF

Research Problem

- Existing IDS in Kubernetes rely on **rule-based detection** (eg, Falco), need efforts to make them adaptive
- Machine Learning-based IDS need **fine-grained runtime data**, but this data is difficult to collect efficiently

Proposed Solution

K-IDS consists of four main components:

1. Data Collection (eBPF)

- Capture system calls (host-level) **using Tetragon**
- Capture network flow (network-level) **using Cilium**
- Collect container-aware metadata (pod, namespace, container ID)

2. Feature Processing

- Transform raw datas into structured
- Aggregate features at container / pod level
- Prepare data for ML models

3. Detection (ML)

- Supervised Model (DT) -> detect known attack patterns
- Hybrid input (syscall + network) for improved detection context

4. Alerting & Response

- Generate alerts from ML detection
- Forward alerts to SIEM

References

Enhancing intrusion detection in containerized services: Assessing machine learning models and an advanced representation for system call data (ScienceDirect: Computers & Security. 2025)		K-IDS: An Intrusion Detection System in Kubernetes Clusters Using eBPF and Machine Learning
- Supervised: DT, RF, SVM, NB - Unsupervised: IF, LOF, One-Class SVM, EE - Deep learning: MLP	- Not yet tested on a live Kubernetes cluster	- Supervised: XGBoost - Unsupervised: LOF, IF (anomaly detection) - Focus: eBPF telemetry, Syscall event, Kubernetes-native, end-to-end IDS

References

Hybrid Runtime Detection of Malicious Containers Using eBPF (ScienceDirect: Computers, Materials and Continua. 2026)

- | | |
|---|--|
| <ul style="list-style-type: none">- Supervised: KNN, Decision Tree, Naive Bayes, RNN, LSTM- Deep learning: MLP | <ul style="list-style-type: none">- Multi-model comparison- Container-based |
|---|--|

References

scenarios. [Fig. 1](#) illustrates the data-processing flow of the proposed architecture, showing how collected system call and network flow data are preprocessed and subsequently passed to a machine learning-based detection module.

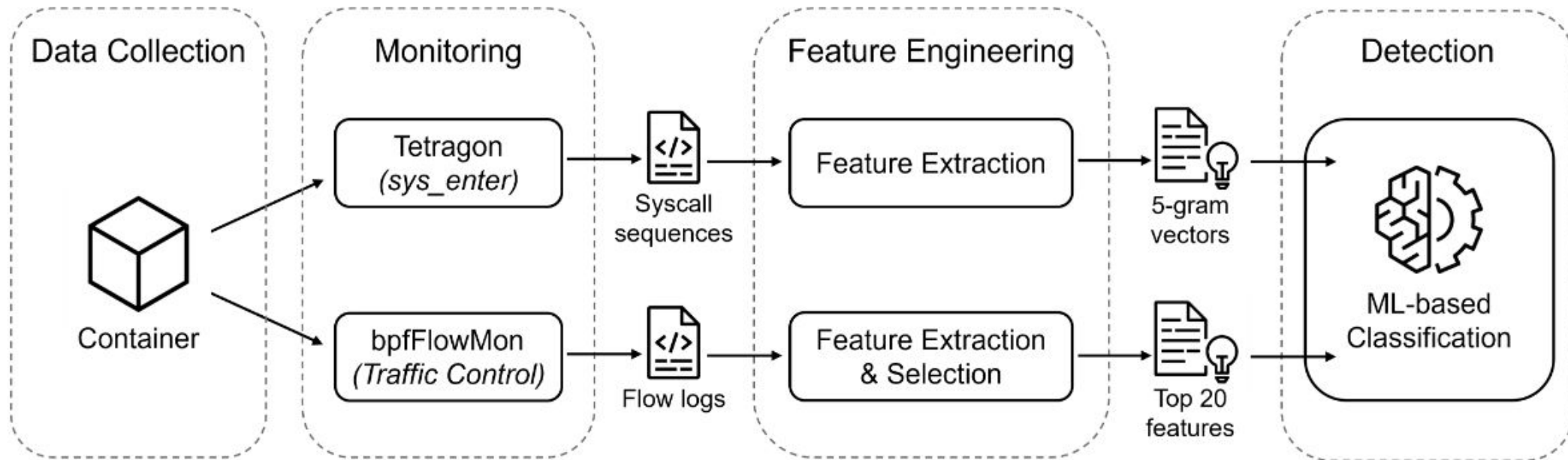
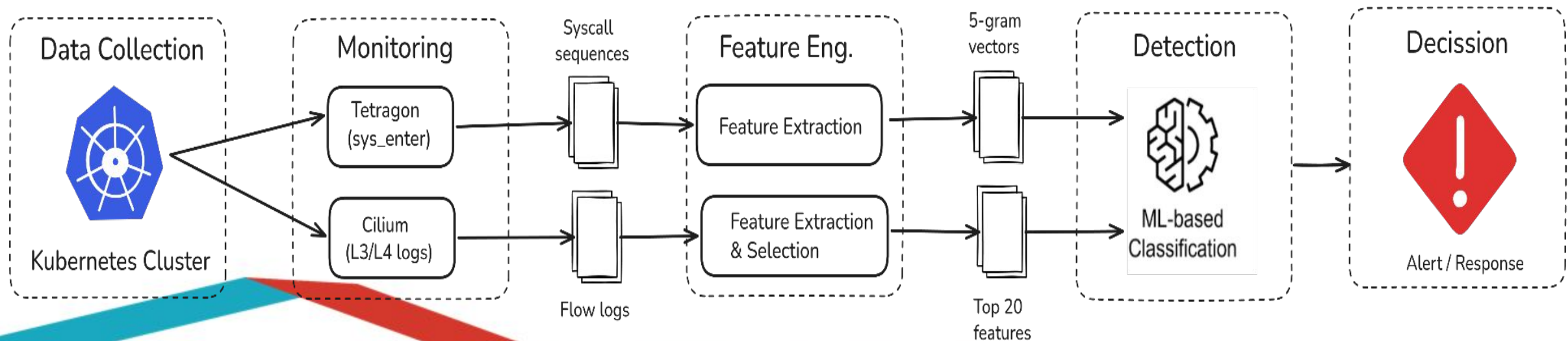
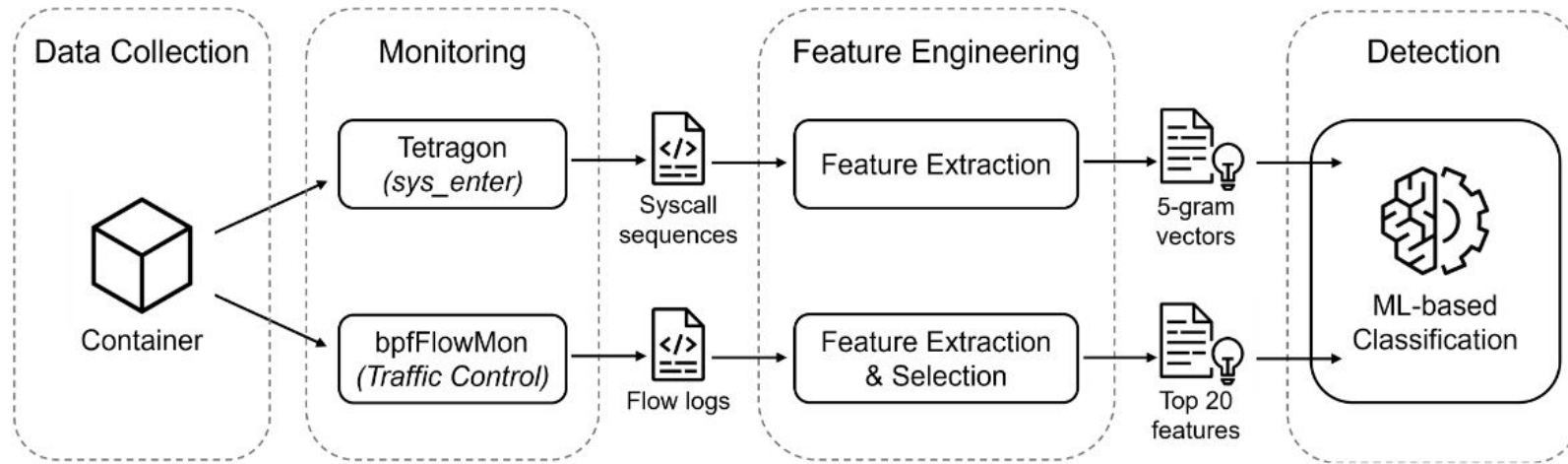
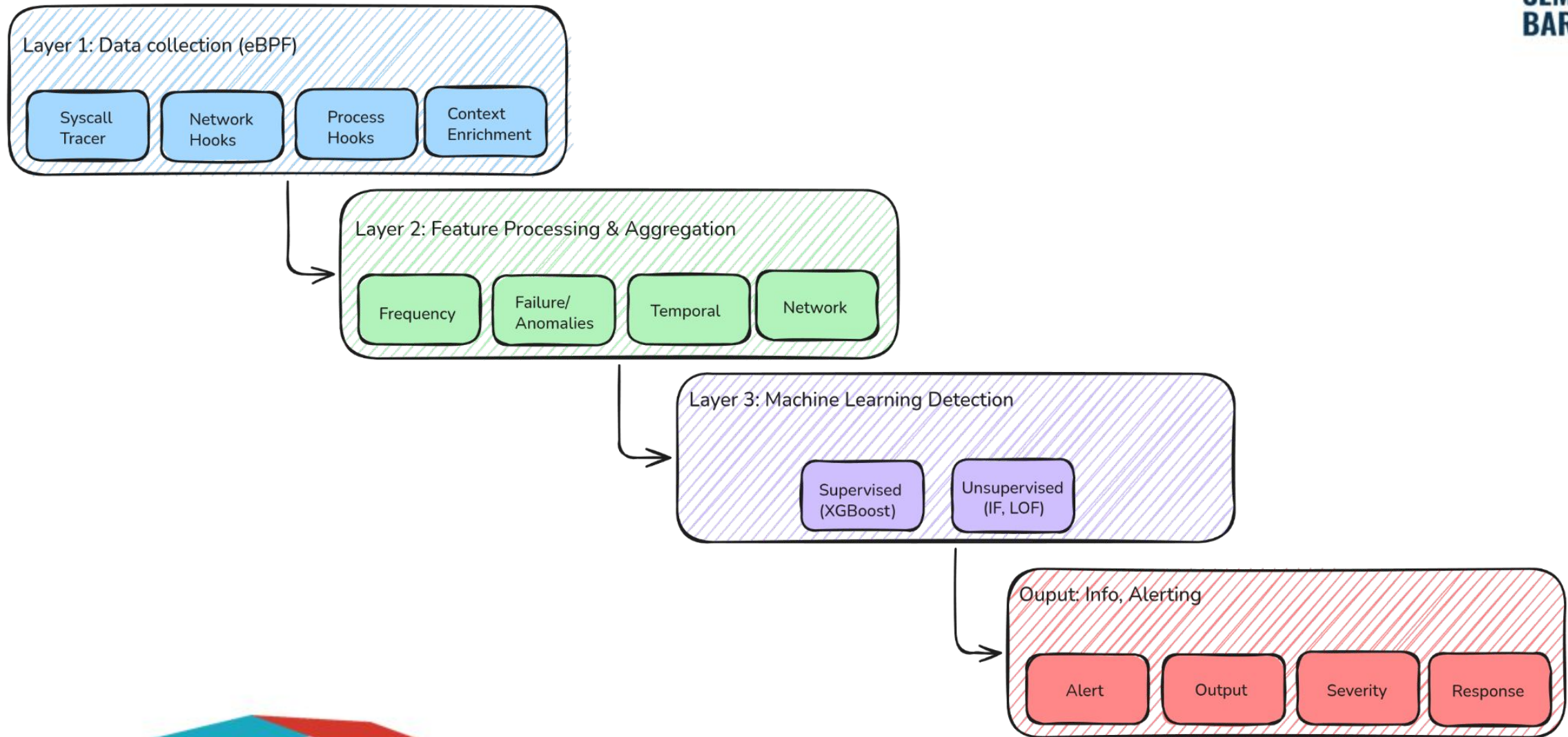


Figure 1: Overview of the proposed framework

Proposed Framework



System Overview



Resources

Dataset	Tools
- A KUBERNETES DATASET FOR MISUSE DETECTION - ADFA-LD (Australian Defence Force Academy Linux Dataset) - Advancing Network Security: A Comprehensive Testbed and Dataset for Machine Learning-Based Intrusion Detection	- eBPF: Tetragon, Cilium - SIEM: Grafana, Loki

Discussions

- Absar Hussain, Abdul Aziz , Hassan Jamil Syed, Shoaib Raza: Preventing IP Spoofing in Kubernetes Using eBPF
- Iwan Handoyo Putro, Tohari Ahmad, Royyana Muslim Ijtihadie: Enhancing MQTT Intrusion Detection in IoT Using Machine Learning and Feature Engineering

Thank You